

Telco Analysis of Churned Customers



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Github Link
https://github.com/akshay-22502/Telco_analysis

Introduction

This report analyzes customer churn for a telecom company using data engineering, business intelligence, and machine learning. From 7,032 customer records, we found a **26.6% churn rate**. The most important churn factors are **contract type, length of service (tenure), and payment method**. A logistic regression model achieved an **84.5% ROC-AUC**, helping identify customers who are likely to leave so the business can take action.

Project Overview

The solution stores data in **CockroachDB**, processes it with **Matillion ETL**, and uses a **star schema** with fact and dimension tables. Raw data from **Amazon S3** is cleaned, organized, and loaded into the warehouse to support fast analysis. **PowerBI** is used to explore churn patterns and customer behavior. **Machine learning** is implemented using **Python (scikit-learn)** with logistic regression chosen for clarity and explainability.

Key Findings

Contract type is the strongest churn predictor. Customers on month-to-month plans churn at 57%, compared with 10% for one-year contracts and 3% for two-year contracts. This represents a 19x difference between the highest and lowest risk customer segments.

Customer tenure shows a clear trend. Most churned customers are concentrated in the 0–12 month range, while retained customers peak in the 60–75 month range. This indicates that early-stage customers are the most vulnerable, whereas long-tenured customers tend to be highly loyal.

Internet service type also shows significant churn differences. Fiber Optic users churn at 42%, which is more than twice the churn rate of DSL users (19%) and five times higher than customers without internet service (8%). This suggests potential issues related to service quality, pricing, or competitive pressure within Fiber Optic offerings.

Payment method is another strong indicator. Customers who pay via Electronic Check churn at 45%, the highest rate of all payment types. In contrast, automated methods such as Bank Transfer and Credit Card have churn rates below 16%, a 2.8x difference. This demonstrates a strong link between payment convenience/automation and customer retention.

Tech support is a key retention lever. Customers without tech support churn at 31%, more than double the 15% churn rate among customers who receive support. Demographics also play a role. Senior citizens churn at 42%, compared to 24% for non-seniors, indicating the need for targeted retention strategies for this demographic.

Data Architecture and ETL Pipeline

The data architecture is built using a star schema, incorporating raw staging tables, mapping tables for value standardization, and analytical dimension and fact tables. The ETL pipeline is orchestrated in Matillion, where data is extracted from S3, loaded into CockroachDB staging tables, and transformed using SQL scripts. These transformations handle missing values in TotalCharges, convert text fields to appropriate data types, generate binary flags for service-related features, and standardize categorical values. The final fact table contains 7,032 customer records with detailed measures covering services, demographics, financial metrics, and churn status, all linked to dimension tables for contract types, genders, internet services, and payment methods.

Machine Learning Model

A logistic regression model was trained on 23 features using balanced class weights to address class imbalance, achieving 73.9% accuracy and an 84.5% ROC-AUC score, indicating strong predictive performance. Feature importance analysis shows that month-to-month contracts (+1.30 coefficient) are the strongest driver of churn, followed by high monthly charges (+0.86), while contract tenure (-1.19) emerges as the most influential retention factor, reinforcing that long-term customers are highly loyal. The model's interpretability allows business stakeholders to clearly understand the reasons behind predictions, supporting more informed and data-driven decisions.

Recommendations based on Models

In the short term, the company should launch a contract migration program to move month-to-month customers to one- or two-year plans, which could reduce churn in this segment by 40–50%. It should also examine Fiber Optic service quality and pricing to address its 42% churn rate. Additionally, incentivizing Electronic Check users to shift to automated payment methods could improve retention by 25–30%.

Medium-term efforts should focus on offering Tech Support and Online Security add-ons to high-risk customers identified by the ML model, potentially reducing churn by 15–20%. Targeted programs for senior citizens and customers with less than 12 months of tenure should provide personalized onboarding and support. This approach helps improve confidence, value perception, and long-term engagement.

In the long run, the company should deploy the ML churn prediction model in production to enable real-time churn risk scoring and automated alerts. Integrating these predictive insights into CRM systems will allow proactive intervention. This ensures timely engagement with at-risk customers and supports sustained retention improvements.